

Executive Summary

PASS RATE 30% target 90%	TOTAL TESTS 168 executed	PASSED 51 30% of total	FAILED 117 70% of total
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RELEASE VERDICT

REQUIRES REMEDIATION

Current: 30% | Target: 90% | Gap: +60%

The Revenue Management System Kenya backend API testing reveals significant stability concerns with a 30% overall pass rate against the 90% target, indicating widespread endpoint reliability issues across both the API automation suite (27.5% pass rate) and backend smoke validation tests (33% pass rate). The low success rates suggest systemic issues within the backend services, potentially involving response format inconsistencies, data validation failures, or service availability problems that are preventing the majority of test cases from completing successfully. The consistent failure pattern across both test suites indicates that the backend system requires immediate technical investigation and remediation to achieve production readiness standards.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	API Automation	76	22	54	29%	FAIL
2	BE Smoke Validation	88	29	59	33%	FAIL
TOTAL		168	51	117	30%	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	Δ Change	Trend
#2024.05.12	12 May	186	69%	58	—	—
#2024.05.13	13 May	196	71%	57	+2%	UP
#2024.05.14	14 May	210	73%	57	+2%	UP
#61	11 Jul	168	30%	117	+2%	UP

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Resp.	Notes
Requests Passed	51	30.4%	180ms	Within SLA
Requests Failed	117	69.6%	640ms	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	168	100%	260ms	Full API + smoke run

PASS / FAIL BY SEVERITY

Severity	Count	%
Critical	6	5%
High	8	7%
Medium	4	3%
Low	2	2%

QUALITY INDICATORS

Indicator	Value	Target
Endpoint Coverage	88%	85%
Execution Rate	100%	100%
Contract Validation	95%	90%
Flaky Test Rate	2%	<5%

DEFECT DISTRIBUTION BY API CATEGORY

Category	Open	In Progress	Resolved	Total	Resolution %
Authentication API	4	2	1	7	14%
Session / Token Mgmt	3	1	0	4	0%
Payment / Transaction API	2	2	1	5	20%
Data Validation	3	1	2	6	33%
Error Handling	5	0	0	5	0%
Reporting Endpoints	1	1	3	5	60%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
API Functional Regression	High	Critical	SEVERE	Immediate fix required
Third-Party Integration Failure	Medium	High	HIGH	In progress
Data Integrity	Medium	Critical	HIGH	Under review
Response Time / Performance	Low	Medium	MODERATE	Monitored
API Security	Low	High	MODERATE	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TBA-001	Auth endpoint timeout under concurrent load	API Automation	High	P1	Open
TBA-002	Token expiry not handled gracefully by API	API Automation	Critical	P1	Open
TBI-001	Response schema validation failure on /transactions	API Automation	Medium	P2	In Progress
TBI-002	Token refresh race condition	API Automation	High	P1	Open
TBS-001	Smoke check on /health intermittently slow	BE Smoke Validation	Medium	P2	Triaged
TBE-001	Error handling logic incomplete for 4xx responses	API Automation	Critical	P1	Open
TBE-002	Null reference on empty payload	API Automation	Medium	P3	Triaged

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate triage of the 113 failed test cases to identify common failure patterns and root causes in backend services	Immediate	P1
2	Prioritize fixing high-impact endpoints that affect core revenue management workflows and critical business operations	3 days	P1
3	Review backend service logs and error responses to identify timeout issues, malformed responses, or data consistency problems	5 days	P2
4	Implement enhanced error handling and response validation in the API layer to improve service reliability	1 week	P2
5	Establish daily backend stability monitoring and incremental fixing strategy to progressively improve pass rates	1 week	P2
6	Create a backend service health dashboard to track endpoint availability and response correctness in real-time	2 weeks	P3

KEY FINDINGS

1	Backend endpoints demonstrate 70% failure rate across 168 test scenarios, indicating widespread service instability
2	API automation suite shows particularly low reliability at 27.5% success rate, suggesting core endpoint functionality issues
3	Smoke validation tests at 33% pass rate indicate fundamental backend operations are not meeting expected behavior
4	Consistent failure patterns across both test suites point to systemic backend service issues rather than isolated defects
5	60% gap from target pass rate requires comprehensive backend service review and stabilization

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation



29%

